

Fast distributed edge coloring by local palette selection

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Highlight. A graph of maximum degree at most a known bound Δ , supplied with a proper vertex 2-coloring, admits a *deterministic proper edge coloring with $O(\Delta)$ colors in $O(1 + \log^* \Delta)$ communication rounds*. No vertex identifiers are needed, and the bound is independent of the number of vertices.

Abstract

We prove the highlight theorem by an elementary construction of expanding sets and two applications of Hall's marriage theorem. Each old edge color specifies a short list of new colors. One side of the bipartition proposes disjoint sublists, and the other side accepts distinct representatives. This reduces a palette of $L\Delta$ colors to $O(\Delta \log L)$ colors. We also prove a sharper $O(\Delta \log L / \log \log L)$ reduction, an explicit $\lfloor 16.5\Delta \rfloor$ final palette, and a direct two-round $O(\Delta^{5/3})$ algorithm. For arbitrary edge lists, a related construction colors lists of size $O(\Delta \log \Delta / \log \log \Delta)$ in three rounds; batching and partial coloring give linear-list and degree-plus-one-list tradeoffs. We give general-graph consequences, a tight one-round lower bound under explicit initialization conventions, and limitations of two particular set-system constructions. All new constructions and reductions are proved, including their numerical estimates. The probabilistic method is used to construct deterministic finite tables, and no efficiency of their local construction is assumed.

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1 Results, model, and perspective

The input to the central result is a bipartite graph *with its two sides supplied*. We call the sides black and white. This information is stronger than a promise of bipartiteness. The result applies to irregular graphs as well as regular graphs; a common upper bound Δ on their degrees is assumed known. The empty graph is immediate, so throughout the proofs we may assume $\Delta \geq 1$.

1.1 Communication and colors

In the synchronous LOCAL model, each vertex can send an arbitrary finite message along each incident edge in each round and perform arbitrary finite local computation between rounds. Each vertex initially knows its degree, its own distinct ports $0, \dots, \deg(v) - 1$, and its side of the supplied bipartition. Opposite-end ports are initially unknown. On general graphs we use distinct identifiers from a universe of polynomial size in n , the number of vertices. For these general-graph upper bounds, n (or a polynomial estimate of it and of the identifier range) may be supplied. The anonymous LOCAL bounds permit any common upper bound Δ . For general-graph and bandwidth statements we assume $1 \leq \Delta \leq n - 1$ on a nonempty simple graph; this holds when Δ is the actual maximum degree. In the general-graph setting a larger supplied bound may be clamped using the supplied n . Unless an intermediate one-sided state is explicitly specified, *both endpoints output the same final color of their edge*. These conventions are especially relevant to exact one- and two-round bounds.

An edge coloring is proper if incident edges receive different colors. In list edge coloring, each edge e has a finite set L_e of permissible colors, initially known at both endpoints. Labels have finite encodings and a common total order; their lengths and their global universe are unrestricted. All list results and the ordinary corollaries derived from them are in LOCAL. The ordinary palette-reduction algorithms and the direct two-round construction also fit CONGEST, where each edge carries $O(\log n)$ bits per round, under the same unrestricted-local-computation convention; see [section 2.1](#). Graphs are finite and simple unless otherwise stated. For a simple graph G , write

$$d_G(e) = \deg_G(u) + \deg_G(v) - 2 \quad (e = uv)$$

for the degree of e in the line graph. Thus a degree-plus-one edge list has size at least $d_G(e) + 1$. We use “degree-plus-one” in this line-graph sense, not to mean $\Delta + 1$.

All tables below are uniform sufficient guarantees, not optimality claims. Logarithms in asymptotic expressions are cut off at an absolute constant. For exact parameter choices, put

$$h = 1 + \log^* \Delta, \quad \ell = 1 + \log^* n, \quad \rho = \min\{r \in \mathbb{N} : r \geq 3, r^{r-2} \geq 720\Delta^4\}. \quad (1)$$

Here $\log^*$ uses base two, and $\rho = \Theta(\log \Delta / \log \log \Delta)$ as $\Delta \rightarrow \infty$. Taking logarithms of its defining inequality proves this last assertion: $(r - 2) \ln r$ must be $\Theta(\ln \Delta)$, and constant multiples of $\ln \Delta / \ln \ln \Delta$ bracket the least feasible r for large Δ .

1.2 Palette size versus time

In particular, 100Δ -element lists take $O(\rho)$ rounds, whereas ordinary $O(\Delta)$ coloring takes $O(h)$ rounds. Choosing $k = \lceil \Delta/h \rceil$ in the batching bound gives lists of size $O(\Delta\rho/h)$ in $O(h)$ rounds. Choosing $k = \lceil \Delta/(1 + \log \Delta) \rceil$ gives

$$2\Delta + O\left(\frac{\Delta}{\log \log \Delta}\right) \quad \text{colors or list entries in } O(1 + \log \Delta) \text{ rounds.} \quad (2)$$

Task with a supplied bipartition	Sufficient palette or list size	Rounds
Ordinary	Δ^2	1
Ordinary	$< 3\Delta^{5/3}$	2
Ordinary	$O\left(\Delta \frac{\log^{(t)} \Delta}{\log^{(t+1)} \Delta}\right)$, any fixed $t \geq 1$	$t + 2$
Ordinary	$\lceil 16.5\Delta \rceil$	$O(h)$
Lists	Δ^2	2
Lists	$\rho(d_G(e) + 1)$ on edge e	3
Lists	$2\Delta - 2 + \rho(2k - 1)$, $1 \leq k \leq \Delta$	$O(h + \Delta/k)$
Lists	$\lceil (2 + \varepsilon)\Delta \rceil$, $\varepsilon > 0$	$O(h + 1 + \rho/\varepsilon)$
Lists	$d_G(e) + 1$ on edge e	$O(\rho(1 + \log \Delta))$

Table 1: Deterministic bounds with a supplied bipartition. Iterated logarithms are denoted by $\log^{(t)}$. The $t + 2$ bound pipelines reductions by alternating the active side, with a final notification to both endpoints.

Task on a general graph	Sufficient size	Total rounds
Ordinary	$3\lceil 16.5\Delta \rceil \leq 50\Delta$	$O(\ell)$
Lists	$O_c(\Delta^2 \log \Delta / 2^{c_h})$, any fixed $c > 0$	$O_c(\ell)$
Lists	100Δ	$O(1 + \log \Delta + \ell)$
Ordinary or lists	$2\Delta + O(\Delta / \log \log \Delta)$	$O(1 + \log \Delta + \ell)$
Lists	$d_G(e) + 1$ on edge e	$O(\rho(1 + \log \Delta) + \ell)$

Table 2: General-graph consequences in the identifier-based LOCAL model. The additive dependence on n is included.

Exact $2\Delta - 1$ colors follow by assigning every edge the same list of that size. Its time is $O(\rho(1 + \log \Delta))$.

The constants in the fixed- t and fixed- c families may depend on those parameters. Consequently these families have no single smallest member if only an unspecified constant round budget is given. The asymptotic expressions have the usual bounded-degree interpretation. Every sufficient nonintegral list size is rounded up.

1.3 Prior ingredients and the main questions

The elementary proof in [section 2](#) uses Hall’s theorem [1] and the probabilistic construction of expanding neighborhoods. This is the standard lossless-expander counting method; a general existence theorem appears in Berinde et al. [4, Proposition 6]. We give the full specialized proof with explicit constants. The sharper ordinary reductions use Haxell’s hypergraph matching condition [2]; its exact bipartite formulation also appears in Annamalai [3, Theorem 1.1]. We include a proof of the needed form.

The proposal-and-acceptance structure is related to the two-round color-reduction and list-coloring framework of Maus and Tonoyan [11]. The list slack-reduction strategy comes from Balliu, Kuhn, and Olivetti [8, Lemma 4.2], also used in Balliu, Brandt, Kuhn, and Olivetti [9, Appendix D]. We prove the version needed here and use virtual vertex splitting to obtain only $O(\rho)$ sequential classes. General-graph reductions use standard vertex-color reduction [5], defective coloring [6, 7], and, for the second row of [table 2](#), the conflict-coloring theorem of Fraigniaud, Heinrich, and Kosowski [10].

For comparison, [9, Lemma 6.1] gives $(2 + \varepsilon)\Delta$ colors in $O(\varepsilon^{-6} \log^{11} \Delta)$ deterministic CONGEST rounds when the bipartition is supplied. At most $2\Delta - 2$ colors have a quite different behavior: Chang, He, Li, Pettie, and Uitto [12, Theorem 2] prove n -dependent lower bounds even with a supplied bipartition. That theorem does not give a lower bound for a

larger constant times Δ . We do not establish whether ordinary $O(\Delta)$ coloring is possible in constant time, or whether arbitrary 100Δ -element lists can be colored in $O(h)$ rounds. The restrictions proved in [section 8](#) concern specified constructions and do not settle these questions.

2 The elementary proof: expansion and Hall twice

Lemma 2.1 (Hall's theorem [\[1\]](#)). *Finite sets A_i , indexed by $i \in I$, admit distinct representatives $a_i \in A_i$ if and only if $|\bigcup_{i \in J} A_i| \geq |J|$ for every $J \subseteq I$.*

Proof. Necessity is immediate. For sufficiency, induct on $|I|$. If a nonempty proper $J \subset I$ has equality, match its sets by induction and delete their union from the other sets. For any collection $K \subseteq I \setminus J$, Hall's inequality for $K \cup J$ implies that at least $|K|$ colors remain, so induction matches those sets too. If there is no such tight J , choose an element of one set, delete that set and element, and use induction: every nonempty collection of the remaining sets originally had at least one unit of slack. The cases $|I| \leq 1$ are immediate. $\square$

Lemma 2.2 (Expanding neighborhoods). *For integers $\Delta \geq 1$ and $r \geq 2$, there are sets $N_i \subseteq [48r\Delta]$, $i \in [2^r\Delta]$, with*

$$|N_i| = 2r + 1, \quad \left| \bigcup_{i \in I} N_i \right| \geq (r + 1) |I| \quad \text{whenever } |I| \leq \Delta.$$

Here $[a] = \{1, \dots, a\}$.

Proof. Choose the N_i independently and uniformly among $(2r + 1)$ -subsets of $[Q]$, where $Q = 48r\Delta$. A fixed D -subset chosen uniformly from $[Q]$ lies in a fixed t -set with probability

$$\frac{\binom{t}{D}}{\binom{Q}{D}} = \prod_{j=0}^{D-1} \frac{t-j}{Q-j} \leq (t/Q)^D$$

when $t \geq D$; it has probability zero otherwise. We use $\binom{a}{b} \leq (ea/b)^b < (3a/b)^b$ for $1 \leq b \leq a$. For completeness, the binomial theorem gives $\binom{a}{b}(b/a)^b \leq (1 + b/a)^a \leq e^b$.

For a fixed $1 \leq k \leq \Delta$, a union of at most $(r + 1)k$ colors is contained in some set of exactly that size. A union bound gives probability at most

$$\begin{aligned} & \binom{2^r\Delta}{k} \binom{48r\Delta}{(r+1)k} \left(\frac{(r+1)k}{48r\Delta} \right)^{(2r+1)k} \\ & \leq \left[9 \left(\frac{r+1}{8r} \right)^r \left(\frac{k}{\Delta} \right)^{r-1} \right]^k \leq (81/256)^k < (1/3)^k. \end{aligned}$$

The second inequality uses $(r+1)/r \leq 3/2$ and $r \geq 2$. Summing over k gives failure probability less than $1/2$. The resulting neighborhoods in fact have strictly more than $(r + 1) |I|$ neighbors for every nonempty eligible I . $\square$

Lemma 2.3 (One palette reduction). *An existing proper edge coloring with at most $2^r\Delta$ colors can be replaced by one with $48r\Delta$ colors in two rounds, for $r \geq 2$.*

Proof. Associate the fixed neighborhood N_i of [lemma 2.2](#) with old color i . At a black vertex, choose an $(r + 1)$ -subset $S_e \subseteq N_{c(e)}$ for every incident edge, with the S_e pairwise disjoint. Such a choice exists: replace each incident edge by $r + 1$ clones whose available colors are $N_{c(e)}$. A set of clones supported on k edges has size at most $(r + 1)k$ and that many available

colors by expansion. Hall matches all clones, giving the required sets. Distinct old colors on a star are essential here.

In the first round, black vertices send their proposals S_e to white vertices. Consider any k edges at a white vertex. Their old colors are distinct and

$$\left| \bigcup_e S_e \right| \geq \left| \bigcup_e N_{c(e)} \right| - \sum_e |N_{c(e)} \setminus S_e| \geq (r+1)k - rk = k. \quad (3)$$

Hall therefore supplies distinct representatives for all received proposals. The white vertex assigns these representatives to its edges and returns them in the second round. They are distinct at white by construction and at black because the proposals were disjoint. $\square$

This proof explains the choice $2r+1$: expansion by $r+1$ allows black vertices to allocate $r+1$ colors per edge, and deleting the other r colors per neighborhood still leaves the one unit of expansion needed at white vertices. The algorithm only needs the neighborhoods and ordinary bipartite matchings.

Theorem 2.4 (Linear palettes in iterated-logarithmic time). *A supplied-bipartite graph of maximum degree at most Δ has a uniform deterministic 512Δ -edge-coloring algorithm taking $O(h)$ rounds. Identifiers are unnecessary.*

Proof. Exchange ports once and color an edge by the ordered pair of its black and white ports. This is a proper Δ^2 -coloring known at both ends. While the current palette has size $L\Delta$ with integer $L > 512$, put $r = \lceil \log_2 L \rceil$ and apply lemma 2.3. The new overhead is $f(L) = 48 \lceil \log_2 L \rceil$. For $r \geq 10$, $48r < 2^{r-1} < L$: the first inequality holds at $r = 10$ and persists since $(r+1)/r < 2$. Thus the integer overhead strictly decreases to at most 512.

For $x \geq 2$, $f(x) \leq 96 \log_2 x$. For all x above an absolute threshold, $f(f(x)) \leq \log_2 x$, since $96 \log_2(96 \log_2 x) \leq \log_2 x$ eventually. Each pair of reductions therefore removes at least one logarithmic level until that threshold; below it there are only an absolute number of decreasing integer steps. This is $O(1 + \log^* \Delta)$ reductions.

The existence argument is uniform: enumerate all possible arrays (N_i) lexicographically and stop at the first satisfying the finite expansion tests. Every vertex uses the same parameters and constructs the same array. Exhaustive search also finds the local matchings. These computations terminate and consume no communication rounds. $\square$

2.1 Alternating sides and message sizes

After the proposal round of lemma 2.3, all current colors are known at the white vertices, and the conceptual coloring is proper everywhere. One may immediately reverse the roles: white vertices choose disjoint proposals for the next reduction, and black vertices accept representatives. A passive vertex need not know old color indices to compute a matching among the received proposals. It only needs the guarantee, supplied by properness of the conceptual old coloring, that those indices are distinct.

Consequently $t \geq 1$ consecutive reductions take t proposal rounds and one final notification. Including the initial port exchange, the total is $t+2$. A reduction viewed in isolation with agreed input and output colors still takes two rounds. This also explains the one-round description in an active/passive formulation of a bipartite problem.

To implement a proposal in CONGEST, index the possible proposals inside each old-color neighborhood. Send the old color and this index; the recipient reconstructs the proposal using the common table. There are $\binom{2r+1}{r+1} \leq 2^{2r+1}$ proposals, so the index needs $O(r)$ bits. Along the reduction sequence $r = O(\log \Delta)$ and every color domain is polynomial in Δ . All messages use $O(\log \Delta)$ bits, hence $O(\log n)$ for simple graphs with a known bound

$\Delta \leq n - 1$. The ordinary constructions also work for loopless bipartite multigraphs with distinct local ports. The relation $\Delta \leq n - 1$ used for the bandwidth statement is specific to simple graphs. For the sharper reduction of [theorem 3.5](#), the candidate count satisfies $\log m = O(\log L + r) = O(\log \Delta)$ along the relevant sequences; in [appendix B](#) it is an absolute constant. The same index encoding therefore applies there as well. This observation addresses bandwidth only; finite-table construction and storage may be extremely expensive.

3 Sharper ordinary palette reductions

Definition 3.1. For an input domain $[M]$ and output domain $[Q]$, a palette system is a nonempty finite family $\mathcal{F}_i$ of nonempty subsets of $[Q]$ for each $i \in [M]$. It is *good up to degree Δ* if, for every $I \subseteq [M]$ of size at most Δ :

1. there are pairwise disjoint choices $S_i \in \mathcal{F}_i$ for $i \in I$;
2. every choice $S_i \in \mathcal{F}_i$ for $i \in I$ admits distinct representatives.

Lemma 3.2 (Palette protocol). *A good palette system converts an M -edge coloring to a Q -edge coloring in two rounds. Its reductions can be pipelined as in [section 2.1](#).*

Proof. Black vertices choose disjoint palettes using the first property and send them. White vertices use the second property to choose distinct representatives and return them. This is the proof of [lemma 2.3](#), with its two Hall arguments replaced by the two stipulated properties. $\square$

The following classical matching condition will be used to obtain the first property. A full proof is given in [appendix A](#).

Theorem 3.3 (Haxell's condition [[2](#), [3](#)]). *Let $\mathcal{F}_i$, $i \in I$, be nonempty finite families of nonempty sets of size at most r . If, for every nonempty $J \subseteq I$, every set meeting every palette in all families indexed by J has size greater than $(2r - 1)(|J| - 1)$, then pairwise disjoint choices $S_i \in \mathcal{F}_i$ exist.*

Lemma 3.4 (Robust Hall estimate). *Independently sample m uniform r -subsets of $[R\Delta]$ for each of $L\Delta$ families, where $r \geq 2$, $R\Delta \geq r$, and the domain sizes are integers. If $a = 9Lm/R^{r-1} < 1$, the probability that the second property of [definition 3.1](#) fails is at most $a^2/(1 - a)$. The same bound holds for ranges of independent uniform r -tuples.*

Proof. A Hall violation gives $s + 1$ different families and one palette from each, all contained in an s -set, for $1 \leq s < \Delta$. To see this, take a violating collection, let s be its union size, and retain $s + 1$ of the indices. Nonempty palettes exclude $s = 0$. For a fixed s the union bound is

$$\begin{aligned} A_s &\leq \binom{R\Delta}{s} \binom{L\Delta}{s+1} [m(s/(R\Delta))^r]^{s+1} \\ &\leq \left[9(L/R)m (s/(R\Delta))^{r-2} \right]^{s+1} \leq a^{s+1}. \end{aligned}$$

In the second line, bound both binomial coefficients by $(3N/k)^k$, increase the exponent s of $3R\Delta/s$ to $s + 1$, and use $s + 1 \geq s$. The containment bound applies both to exact subsets and to tuple ranges. Summing the geometric series proves the claim. $\square$

Theorem 3.5 (Sharper reduction). *For integers $L \geq 6$ and $\Delta \geq 1$, let $r \geq 3$ be the least integer with $r^r \geq 128L^2$. A proper $L\Delta$ -edge coloring can be reduced in two rounds to $8r\Delta$ colors. In particular, the overhead becomes $O(\log L / \log \log L)$.*

Proof. Put

$$R = 8r, \quad q = (3/4)^r, \quad m = \lceil 12(L+r)/q \rceil.$$

For each old color sample m independent uniform exact r -subsets of $[R\Delta]$. First exclude covers of all palettes in any k families by a set T of size $2r(k-1)$, for $2 \leq k \leq \Delta$. Writing $x = k/\Delta$, the probability that one sample meets this fixed T is at most

$$g(x) = 1 - (1 - x/4)^r.$$

Indeed, the exact avoidance probability is $\prod_{j=0}^{r-1} (1 - 2r(k-1)/(R\Delta - j))$, and each fraction is at most $2rk/(R\Delta) = x/4$, since $R\Delta \geq kj$.

The logarithmic elasticity $xg'(x)/g(x)$ decreases on $(0, 1]$: with $y = 1 - x/4$ it equals $r/(1 + y^{-1} + \dots + y^{-(r-1)})$. Therefore, with

$$\alpha = g'(1)/g(1) = \frac{rq}{3(1-q)}, \quad g(x) \leq (1-q)x^\alpha.$$

The last inequality follows by integrating $d \ln g(x)/d \ln x \geq \alpha$ from x to 1. Monotonicity of $z \ln(3R\Delta/z)$ on $0 < z \leq R\Delta$ gives

$$\binom{R\Delta}{2r(k-1)} \leq \left(\frac{3R\Delta}{2rk} \right)^{2rk}.$$

Thus the cover-witness probability for size k is at most

$$B_k \leq [b x^{m\alpha - 2r - 1}]^k \leq b^k, \quad b = 3L 12^{2r} (1-q)^m,$$

because $m\alpha \geq 4r(L+r)/(1-q) > 2r+1$. Using $mq \geq 12(L+r)$, $e > 2$, and $L \leq 2^L$,

$$b \leq 3L 12^{2r} e^{-12(L+r)} \leq 3L 2^{-12L} (9/256)^r < 1/4.$$

The sum of the cover probabilities is at most $b^2/(1-b) < 1/12$. Any cover of size at most $(2r-1)(k-1)$ can be padded to size $2r(k-1)$; all these sizes are below $R\Delta$. Singleton families have positive hitting number. Hence absence of the witnesses implies Haxell's condition for every star.

For robust Hall, [lemma 3.4](#) applies with $a = 9Lm/(8r)^{r-1}$. The ceiling gives $m \leq 13(L+r)/q$, so

$$a \leq \frac{156L(L+r)}{(6r)^{r-1}}.$$

The integer L itself qualifies in the definition of r because $L^{L-2} \geq 6^4 \geq 128$; thus $r \leq L$. Also $6^{r-1}/r \geq 12$ for $r \geq 3$, by induction. Consequently

$$a \leq \frac{312L^2}{12r^r} \leq \frac{13}{64} < \frac{1}{4}.$$

Both failure sums together are less than $1/6$, proving existence of a good system. Canonical finite search and [lemma 3.2](#) give the algorithm. Taking logarithms of $r^r \geq 128L^2$ proves the claimed asymptotic size of the least r . $\square$

Corollary 3.6 (Fixed-round hierarchy). *For every fixed integer $t \geq 1$, a supplied-bipartite graph can be edge colored using*

$$O\left(\Delta \frac{\log^{(t)} \Delta}{\log^{(t+1)} \Delta}\right) \quad \text{colors in } t+2 \text{ rounds.}$$

Proof. Start with the one-round port-pair coloring and apply [theorem 3.5](#) t times, using [section 2.1](#). For fixed t and sufficiently large Δ , composition of $O(\log x / \log \log x)$ has the displayed bound; this follows inductively from $\log(\log x / \log \log x) = \Theta(\log \log x)$ and the corresponding next logarithm. Bounded exceptional degrees can retain the original port-pair coloring, with a larger constant depending on t . $\square$

Theorem 3.7 (An explicit linear coefficient). *There is a deterministic supplied-bipartite algorithm using $\lfloor 33\Delta/2 \rfloor$ colors in $O(h)$ rounds.*

Proof. The complete numerical construction in [appendix B](#) reduces 512Δ colors through the coefficients

$$512 \longrightarrow 100 \longrightarrow 20 \longrightarrow 18 \longrightarrow 17 \longrightarrow 16.6 \longrightarrow 16.5$$

in a constant number of reductions for every $\Delta \geq 6000$. Combine it with [theorem 2.4](#). For $\Delta < 6000$, process the 512Δ old color classes sequentially. Each is a matching; give each edge a color from $[2\Delta - 1]$ avoiding previously assigned colors. At most $2\Delta - 2$ colors are forbidden. A constant number of rounds per class suffices, and the number of classes is bounded by an absolute constant in this branch. This gives the assertion for all degrees. $\square$

4 A direct algorithm for exactly two rounds

The preceding reductions need an initial proper coloring. The following algorithm works directly from local ports and uses only grouping and ranks.

Theorem 4.1. *For a supplied-bipartite graph of maximum degree at most $\Delta \geq 1$, there is a deterministic two-round edge-coloring algorithm using*

$$\Delta(2s + b^2) < 3\Delta^{5/3} \quad \text{colors,} \quad s = \left\lfloor \Delta^{2/3} \right\rfloor, \quad b = \lfloor \Delta / (s + 1) \rfloor.$$

Both endpoints know the output after the second round, and messages have $O(\log \Delta)$ bits. The construction also works with parallel edges.

Proof. Exchange ports in round one. For edge e , let $p(e)$ be its black port and $q(e)$ its white port, and put $z(e) = (p(e) + q(e)) \bmod \Delta$. At a black vertex, group edges by their opposite port q . Call a group light if it has at most s edges and heavy otherwise. There are at most b heavy groups. Give every light edge a zero-based rank $a_B(e) < s$ within its group, in increasing order of p . Give a heavy edge a zero-based rank $b_B(e)$ among *all black-heavy edges with the same z* , again in port order. This rank is less than b : for a fixed z and opposite port q , there is at most one local port $p \in [0, \Delta - 1]$. Thus each heavy group contributes at most one edge at that z . White vertices compute the symmetric flags and ranks a_W, b_W . These computations use only information received in round one; heavy ranks do not depend on the other endpoint's classification.

In round two each endpoint sends its light/heavy flag and the applicable rank. Both endpoints then use the same priority rule:

$$c(e) = \begin{cases} (B, q(e), a_B(e)), & \text{if black calls } e \text{ light,} \\ (W, p(e), a_W(e)), & \text{otherwise, if white calls } e \text{ light,} \\ (H, z(e), b_B(e), b_W(e)), & \text{otherwise.} \end{cases}$$

The tags specify disjoint output palettes. Within tag B , the pair (q, a_B) distinguishes edges at black, while the local port q distinguishes them at white. Tag W is symmetric. Within tag H , different z values differ, and at fixed z the appropriate local heavy rank distinguishes

incident edges at each endpoint. Restricting those ranks to edges heavy at both endpoints preserves distinctness. Hence the coloring is proper and both endpoints compute the same result.

The tags use at most Δs , Δs , and Δb^2 colors. Since $s \leq \Delta^{2/3}$ and $s + 1 > \Delta^{2/3}$, we have $b < \Delta^{1/3}$, proving the strict displayed bound. If $b = 0$, the heavy tag is unused. Each message contains a port, or a flag and a rank, and therefore has $O(\log \Delta)$ bits. $\square$

5 Three-round coloring from arbitrary lists

Ordinary palette reduction does not directly preserve edge lists: an allowed list may avoid every predetermined candidate palette. We replace disjoint-choice families by families that are rich inside every sufficiently large allowed set. We first work in a finite common universe, and then remove that assumption by a local encoding.

Lemma 5.1 (Rich families with robust representatives). *Let $r \geq 3$, $C \geq 1$, and $r^r \geq 180C^2$. For every integer $j \geq 1$ with $rj \leq C$ there are j families $\mathcal{F}_{j,1}, \dots, \mathcal{F}_{j,j}$ of nonempty subsets of $[C]$, each of size at most r , such that:*

1. *every rj -subset of $[C]$ contains a palette from each family;*
2. *every selection of palettes from distinct families admits distinct representatives.*

Proof. Put $t = rj$ and $m = \lceil 4C(C/t)^r \rceil$. In each family, independently sample m independent uniform r -tuples in $[C]$ and use their ranges as palettes. A fixed t -set contains the range of a fixed tuple with probability $(t/C)^r$. Thus a union bound for richness failure gives

$$j \binom{C}{t} (1 - (t/C)^r)^m \leq C 2^C e^{-4C} < C 2^{-3C} \leq 1/8.$$

For robust Hall, a violation is witnessed by $s + 1$ different families with a palette each inside an s -set, for $1 \leq s < j$. Its probability is at most

$$\begin{aligned} B_s &\leq \binom{C}{s} \binom{j}{s+1} m^{s+1} (s/C)^{r(s+1)} \\ &\leq \left[\frac{9jms^{r-2}}{C^{r-1}} \right]^{s+1} \leq a^{s+1}, \quad a = \frac{9jms^{r-2}}{C^{r-1}}. \end{aligned}$$

The binomial calculation is the one in [lemma 3.4](#). Since $m \leq 5C(C/(rj))^r$,

$$a \leq \frac{45C^2}{r^r j} \leq \frac{1}{4j} \leq \frac{1}{4}.$$

Summing gives Hall-failure probability at most $1/12$. There are no Hall witnesses when $j = 1$. The total failure probability is below one, so each required j -table exists. It is constructed separately for each j ; no union bound over different values of j is needed. The properties are finite and decidable, hence canonical exhaustive search gives a common deterministic table. $\square$

Lemma 5.2 (Finite-universe list coloring). *Under the assumptions $r \geq 3$ and $r^r \geq 180C^2$, a supplied-bipartite graph whose lists satisfy*

$$L_{uv} \subseteq [C], \quad |L_{uv}| \geq r(\deg(u) + \deg(v) - 1)$$

can be list edge colored in three rounds.

Proof. In round one every white vertex sends its degree j and its one-based port index $i \in [j]$ along each incident edge. The appropriate j -table exists: on every edge at that white vertex, the list-size promise implies $rj \leq C$.

A black vertex of degree b processes its edges in local port order. For an edge going to white degree j , remove the colors in all its previously chosen palettes. At most $r(b-1)$ colors have been removed, leaving at least $r(b+j-1) - r(b-1) = rj$ colors. Richness supplies a palette from $\mathcal{F}_{j,i}$ contained in the remainder. In round two send that palette to white. All palettes chosen at a black vertex are disjoint, even if its incident family indices repeat.

At a white vertex the family indices are its distinct ports in one fixed j -table. Robust Hall gives distinct representatives. Choose the lexicographically first such assignment and return the colors in round three. They are allowed by the lists and distinct at both sides. $\square$

Theorem 5.3 (Arbitrary labels, three rounds). *Let ρ be defined by (1). If the lists on a supplied-bipartite graph of maximum degree at most Δ satisfy*

$$|L_{uv}| \geq \rho(\deg(u) + \deg(v) - 1),$$

then they can be colored in three deterministic rounds, with no bound on the global color universe.

Proof. Set $K = 2\rho\Delta$ and $C_0 = 2\rho\Delta^2$. Truncate each edge list canonically to its first $\min\{|L_e|, K\}$ colors; both endpoints perform the same truncation. Its pointwise promise survives because $\rho(\deg(u) + \deg(v) - 1) < K$. At a white vertex w , form the union U_w of its truncated incident lists. This union has size at most C_0 . Encode it injectively in $[C_0]$ by its order ranks, writing ϕ_w for the encoding. The definition of ρ gives

$$\rho^\rho \geq 720\rho^2\Delta^4 = 180C_0^2.$$

Thus lemma 5.1 supplies all necessary tables on $[C_0]$.

In round one w sends the ordered union U_w , its degree, and its port along each incident edge. A black vertex now knows the encoding and inverse for each of its edges. It chooses palettes greedily as in lemma 5.2, but removes colors using their *actual labels*. For the current edge, encode an available ρj -set using the appropriate ϕ_w , choose a palette inside that image by richness, and decode it to actual labels. Send the actual palette in round two. No unused coordinate is chosen because the palette is contained in an image of actual allowed colors.

In round three each white vertex encodes its received palettes using its single map ϕ_w , chooses distinct representatives using its table, and sends back their actual labels. Injectivity preserves distinctness at white. At black, disjointness was enforced on actual labels across all incident edges, including those using different encodings. This proves correctness and the exact round count. $\square$

One shared encoding per white star is crucial. Independent rank encodings of individual edge lists would not preserve whether two incident actual colors are equal. The size of the messages here is unrestricted; this is a LOCAL result.

6 List slack, batching, and partial coloring

6.1 Batching with constant additive slack

Proposition 6.1. *Lists of size at least Δ^2 can be edge colored in two rounds on a supplied-bipartite graph.*

Proof. Each black vertex chooses disjoint Δ -element sublists for its edges greedily. At most $\Delta(\Delta - 1)$ colors are excluded before any choice, so this is possible. Send them in round one. A white vertex receives at most Δ sets, all of size Δ , so every nonempty subcollection has union size at least its cardinality. Hall gives distinct representatives; return them in round two. $\square$

Theorem 6.2 (Batching tradeoff). *For $1 \leq k \leq \Delta$, lists of size*

$$2\Delta - 2 + \rho(2k - 1) \tag{4}$$

suffice in $O(h + \Delta/k)$ rounds on supplied-bipartite graphs. More generally, lists of size $\lceil (2 + \varepsilon)\Delta \rceil$ suffice in $O(h + 1 + \rho/\varepsilon)$ rounds for every $\varepsilon > 0$.

Proof. Compute a proper 512Δ auxiliary edge coloring using [theorem 2.4](#). Group its colors into consecutive batches of at most k . A batch induces maximum degree at most k . Process batches sequentially, removing colors used on previously colored adjacent edges. An edge has at most $2\Delta - 2$ adjacent edges in total, so (4) leaves at least $\rho(2k - 1)$ colors. Apply [theorem 5.3](#) to the batch, keeping the ambient parameters Δ, ρ fixed. Its pointwise list requirement is no larger than that remainder. Between calls, endpoints exchange the sets of colors already used at their vertices, so both endpoints agree on the residual list. This takes a constant number of rounds per batch. Properness within a batch is supplied by the primitive, and across batches by deleting previously used colors. There are $O(\Delta/k)$ batches.

For the second assertion put $x = \varepsilon\Delta/(2\rho)$. If $x \geq 1$, take $k = \min\{\Delta, \lfloor x \rfloor\}$. At least $\varepsilon\Delta + 2$ colors remain on each edge, which is enough since $\rho(2k - 1) \leq 2\rho k \leq \varepsilon\Delta$. When $1 \leq x < \Delta$, $\lfloor x \rfloor \geq x/2$, hence the number of batches is $O(1 + \rho/\varepsilon)$; the same bound holds when $x \geq \Delta$. If $x < 1$, process matching classes directly, choosing any remaining color. There are $O(\Delta) = O(\rho/\varepsilon)$ classes. $\square$

The simpler quadratic-list primitive can similarly replace $\rho(2k - 1)$ in (4) by k^2 . Thus, if useful for particular finite parameters, $2\Delta - 2 + \min\{k^2, \rho(2k - 1)\}$ also suffices with the same asymptotic time. Setting $\varepsilon = 98$ proves the 100Δ -list bound in [table 1](#). The choices of k described after that table give its other consequences.

6.2 Degree-plus-one lists by halving residual degrees

We give a full slack-reduction proof in the style of [8, Lemma 4.2]. The improvement in the number of sequential classes comes from the ordinary coloring theorem and virtual vertex splitting.

Theorem 6.3 (Partial coloring and degree-plus-one lists). *On a supplied-bipartite graph, suppose $|L_e| \geq d_G(e) + 1$. There is a phase taking $O(\rho)$ rounds which preserves this invariant and reduces the maximum residual edge degree by at least a factor of two, unless all edges are colored. Consequently all edges can be colored in $O(\rho(1 + \log \Delta))$ rounds. More generally t phases cost $O(\rho t)$ rounds and leave maximum edge degree at most $(2\Delta - 2)/2^t$.*

Proof. Let H be the residual graph at the start of a phase and freeze $D_e = d_H(e)$. Set $p = 4\rho$. At every vertex, group its incident residual edges in local port order into blocks of at most p , and replace each block by a virtual vertex with the same bipartition side. The resulting virtual graph has maximum degree at most p and the same edges. One physical vertex simulates its blocks, and each virtual edge uses its corresponding physical edge. Thus one virtual round costs one physical round.

Apply [theorem 2.4](#) with degree bound p , obtaining at most $512p$ auxiliary edge colors in $O(1 + \log^* p)$ rounds. Use these colors as classes on the original graph. A class contains at

most one edge from each block. If $e = uv$ belongs to that class, its edge degree within the class is at most

$$\begin{aligned} \lceil \deg_H(u)/p \rceil - 1 + \lceil \deg_H(v)/p \rceil - 1 &= \lfloor (\deg_H(u) - 1)/p \rfloor + \lfloor (\deg_H(v) - 1)/p \rfloor \\ &\leq D_e/p. \end{aligned}$$

Process the classes in order. Delete colors already used by colored adjacent edges, and declare an edge active if its remaining list has more than $D_e/2$ colors. Endpoints synchronize these lists and the active status in a constant number of rounds. In the active subgraph, the current edge degree D'_e is at most $D_e/(4\rho)$, hence $|L_e| > 2\rho D'_e$. If $D'_e \geq 1$, this implies $|L_e| \geq \rho(D'_e + 1)$, so [theorem 5.3](#) colors the active nonisolated edges in three rounds. Isolated active edges may have shorter lists; color them directly from any available color. They are separate components and can be handled concurrently.

An edge left uncolored was inactive when its class was processed. Its list started with at least $D_e + 1$ colors and then had at most $D_e/2$. Therefore at least $D_e/2 + 1$ colors had been deleted, which requires at least that many adjacent edges to have been colored. Its final residual degree is at most $D_e/2 - 1$. At every stage, coloring an adjacent edge removes one adjacency and deletes at most one list color, preserving the degree-plus-one invariant. In particular, an isolated residual edge is active and will be colored. The maximum residual edge degree halves, and after $O(1 + \log \Delta)$ phases no edge remains.

Each of $512p = O(\rho)$ classes costs constant time, and $1 + \log^* p = O(\rho)$. The phase time and the partial-coloring claim follow. All degree and list information used in a class concerns the current subgraph; the ambient parameters Δ, ρ remain valid. $\square$

7 General graphs

Whenever an algorithm is applied to a subgraph, its incident edges are renumbered locally from zero in their inherited port order; the algorithms use these subgraph ports.

We now consider simple graphs with identifiers, a known maximum-degree bound Δ , and arbitrary finite edge lists known at both endpoints. Write $h = 1 + \log^* \Delta$ and $\ell = 1 + \log^* n$. The parameter $\rho = \rho(\Delta)$ of [Theorem 5.3](#) is kept fixed when we pass to subgraphs. We first explain the ordinary-coloring reduction, then give list-preserving reductions. In particular, multiplying an ordinary color by a refinement color does not preserve arbitrary lists.

7.1 Ordinary coloring by splitting vertices

Theorem 7.1. *A deterministic LOCAL algorithm properly edge-colors a graph of maximum degree at most Δ with $O(\Delta)$ colors in $O(\ell)$ rounds. More precisely, a supplied-bipartite $q(\Delta)$ -edge-coloring algorithm taking $T(\Delta)$ rounds yields a general-graph algorithm using $3q(\Delta)$ colors in $O(T(\Delta) + \ell)$ rounds.*

Proof. Orient every edge from its smaller-identifier endpoint to its larger-identifier endpoint. Replace each vertex v by a black copy v^+ and a white copy v^- , and replace $u \rightarrow v$ by u^+v^- . The resulting virtual graph has maximum degree at most Δ . A physical vertex simulates its two copies, so one virtual communication round costs one physical round. Apply the supplied-bipartite algorithm and project the resulting colors onto the original edges.

For every projected color, each vertex has at most one incoming and at most one outgoing edge of that color. Its monochromatic components are directed paths: a cycle would be directed, contradicting the strictly increasing identifiers along directed edges. A standard deterministic constant-degree coloring algorithm properly 3-edge-colors all these paths in parallel in $O(\ell)$ rounds. Explicitly, host the process for each path edge at its tail, using its tail

identifier as its name. These names are distinct within each path. Apply the proper vertex-coloring algorithm proved in Appendix C to the line graph of each path, whose maximum degree is two, to obtain a constant-size proper palette. Process those color classes sequentially and recolor into [3], avoiding the at most two previously assigned neighbor colors. The simulation has constant cost per line-graph round. Each edge belongs to one path instance, and all instances can be simulated concurrently. Output the pair consisting of the projected color and the path color. Edges with different first coordinates differ, and adjacent edges with the same first coordinate differ in the second coordinate. The first claim now follows from Theorem 2.4, since $h = O(\ell)$. $\square$

7.2 Two standard vertex-coloring tools

The list reductions use the following published distributed vertex-coloring tools. We state their needed quantitative forms here and give complete proofs in Appendix C.

Lemma 7.2 (Standard color reduction and defective coloring). *There is an absolute constant A with the following properties.*

1. *A graph of maximum degree $D \geq 1$ with a proper M -vertex-coloring can obtain a proper AD^2 -vertex-coloring in $O(1 + \log^* M)$ rounds. In particular, identifiers of polynomial range in n give such a coloring in $O(\ell)$ rounds.*
2. *For every integer b with $1 \leq b \leq D$, the same input admits a vertex-coloring with $O((D/b)^2 \log(2D/b))$ colors for which every monochromatic subgraph has maximum degree at most b , in $O(1 + \log^* M)$ rounds. The implied constant in the running time is independent of b .*

The first part is Linial's color reduction [5]. For the second, our proof follows the defective-color-reduction method of Kuhn [6]; the stronger published palette bound omits the logarithmic factor. See also [7]. The slightly weaker palette here has no effect on any of our edge-coloring bounds. Starting from a proper $O(\Delta^2)$ -coloring, both tools take $O(h)$ rounds on every subgraph. We retain that original proper coloring throughout.

Lemma 7.3 (Binary decomposition). *Suppose vertices have labels in $[m]$, not necessarily forming a proper coloring. The edges with distinct endpoint labels can be partitioned into at most $\lceil \log_2 m \rceil$ supplied-bipartite subgraphs in one round. Edges whose endpoint labels agree are omitted.*

Proof. Encode the labels by binary strings of a common length. Assign an edge with different endpoint labels to the first bit position in which its labels differ. At that position the bit supplies the black/white bipartition. Every eligible edge has exactly one such first position. Exchanging labels lets both endpoints determine the edge's part. $\square$

7.3 Degree-plus-one lists

Theorem 7.4. *If every edge uv has a list of size at least $\deg(u) + \deg(v) - 1$, then the graph can be list edge-colored in*

$$O(\rho(1 + \log \Delta) + \ell) = O\left(\frac{\log^2 \Delta}{\log \log \Delta} + \ell\right)$$

rounds, with bounded-logarithm conventions for small Δ .

Proof. Initially obtain a proper $O(\Delta^2)$ -vertex-coloring in $O(\ell)$ rounds. At all times delete the colors of already colored adjacent edges from every uncolored edge's list. If H is the current residual graph, the invariant

$$|L_e| \geq \deg_H(u) + \deg_H(v) - 1$$

holds: coloring one adjacent edge removes one adjacency and deletes at most one available color.

Consider a stage whose residual maximum-degree bound is D . For $D \geq 2$, Lemma 7.2 gives a vertex-coloring with at most c colors and monochromatic degree at most $\lfloor D/2 \rfloor$ in $O(h)$ rounds, where c is an absolute constant. Process each pair of distinct vertex colors in turn. Its uncolored edges form a graph with a supplied bipartition and inherit the degree-plus-one list invariant. Apply $t = \lceil \log_2(16c) \rceil$ phases of the partial coloring algorithm from Theorem 6.3. This costs $O(\rho)$ rounds and leaves maximum edge degree at most

$$\frac{2D - 2}{2^t} \leq \frac{D}{8c}.$$

In any graph with maximum edge degree B , every nonisolated vertex has degree at most $B + 1$, because any edge incident to it has at least $\deg(v) - 1$ adjacent edges. Thus, when $D \geq 8c$, each processed pair leaves maximum vertex degree at most $D/(4c)$. Later pair calls can only remove further edges. Every vertex belongs to at most $c - 1$ pairs, so all the remaining bichromatic edges together have maximum degree at most $D/4$. The unprocessed monochromatic edges have maximum degree at most $D/2$. The residual maximum degree is therefore at most $3D/4$.

Update the common bound to $D \leftarrow \lfloor 3D/4 \rfloor$, which is valid since actual degrees are integers, and repeat until this bound is less than $8c$. There are $O(1 + \log \Delta)$ stages, each costing $O(h + \rho) = O(\rho)$ rounds. For the final graph, reduce the retained proper vertex-coloring to a constant-size proper vertex-coloring in $O(h)$ rounds. This gives a constant-size proper auxiliary edge-coloring: orient edges by vertex color, and label an edge by its two endpoint colors and its port numbers at both endpoints. Edges sharing an endpoint have different labels. Greedily process its constantly many matching classes. The invariant supplies an available color for every edge. All chosen colors belong to the original lists.

This reduction is the list-preserving defective-coloring decomposition used by Balliu, Brandt, Kuhn, and Olivetti [9, Theorem D.4], with the stronger partial-coloring routine proved here. The explicit $O(h)$ decomposition cost is absorbed by ρ . $\square$

7.4 Nearly 2Δ colors in logarithmic time

Theorem 7.5. *For $1 \leq k \leq \Delta$, lists of size*

$$L(k) = 2\Delta - 2 + \rho(2k - 1)$$

suffice in

$$O\left(\ell + \frac{\Delta}{k} + h \log \frac{2\Delta}{k} + \log(2\Delta)\right)$$

rounds on general graphs. Consequently, lists of size

$$2\Delta + O\left(\frac{\Delta}{\log \log \Delta}\right)$$

suffice in $O(\log(2\Delta) + \ell)$ rounds. In particular, for every fixed $\varepsilon > 0$, lists of size $\lceil (2 + \varepsilon)\Delta \rceil$ suffice within this time bound. The same assertions hold for ordinary edge-coloring by using a common list on all edges.

Proof. Obtain and retain a proper $M = O(\Delta^2)$ -vertex-coloring in $O(\ell)$ rounds. At all times an uncolored edge has at least $\rho(2k - 1)$ available colors, since at most $2\Delta - 2$ adjacent edges can ever be colored.

As long as the current maximum-degree bound D exceeds k , compute the constant-color defective vertex-coloring with monochromatic degree at most $\lfloor D/2 \rfloor$ given by Lemma 7.2. Its cost is $O(h)$. Process the constantly many pairs of distinct vertex colors sequentially. For a pair, compute a proper $O(D)$ auxiliary edge-coloring by Theorem 2.4, and group these auxiliary colors into batches of at most k . Theorem 5.3 colors each batch in constant time, since its maximum degree is at most k . Endpoints exchange colors used at their vertices between successive calls to keep their residual lists identical. A whole stage costs $O(h + D/k)$ rounds and leaves only monochromatic edges, hence maximum degree at most $\lfloor D/2 \rfloor$.

There are $O(1 + \log(\Delta/k))$ such stages, and the sum of their degree bounds is $O(\Delta)$. Their total cost is

$$O\left(h \log \frac{2\Delta}{k} + \frac{\Delta}{k}\right).$$

Once the residual degree is at most k , apply Lemma 7.3 to the retained proper M -vertex-coloring. It partitions *all* residual edges into $O(\log(2\Delta))$ supplied-bipartite subgraphs, each of maximum degree at most k . Color these sequentially with the three-round primitive. This proves the stated general bound without any further degree decomposition.

For $\Delta \geq 2$, set $k = \lceil \Delta / \log_2 \Delta \rceil$. Then $1 \leq k \leq \Delta$, $\Delta/k = O(\log \Delta)$, and $h \log(2\Delta/k) = O(h \log \log \Delta) = O(\log \Delta)$. Moreover,

$$L(k) = 2\Delta - 2 + \rho \left(2 \left\lceil \frac{\Delta}{\log_2 \Delta} \right\rceil - 1 \right) = 2\Delta + O\left(\frac{\Delta}{\log \log \Delta}\right).$$

The degree-one case is immediate. For fixed ε , the displayed slack is at most $\varepsilon\Delta$ for sufficiently large Δ . The finitely many smaller degree bounds are handled by a constant-degree greedy schedule after color reduction; its constant may depend on ε . $\square$

7.5 Short-time general-graph list tradeoffs

For completeness we give a family of sufficient list sizes for a total $O(\ell)$ -round algorithm. It uses the following consequence of conflict coloring, whose complete proof is included in Appendix D.

Lemma 7.6 (A consequence of local conflict coloring). *There is an absolute constant B such that a graph of maximum degree $D \geq 1$, supplied with a proper s -vertex-coloring, can be list-colored in $O(1)$ rounds if every list has at least*

$$\max\{s, D, BD^2 \log(2D)\}$$

colors. The labels of these colors may be arbitrary finite strings.

This is the list-coloring specialization of the method of Fraigniaud, Heinrich, and Kosowski [10, Lemma 3.3]. Appendix D proves it directly, using two transformations of conflict-coloring instances and a finite lookup table. Its local relabelings preserve conflicts, so the original labels need no common bounded universe. Degree zero is immediate. At degree one, components are isolated vertices or single edges, and one exchange of lists and proper auxiliary colors permits the smaller-color endpoint to choose first and the other to avoid its choice. When applied to a line graph, each round has constant simulation cost in the original graph: host each edge process at a deterministically selected endpoint, communicate through the shared endpoint of adjacent edges, and finally notify both endpoints of the output.

Theorem 7.7. *For every fixed real $a > 0$, there is a constant C_a such that lists of size at least*

$$\left\lceil C_a \frac{\Delta^2 \log(2\Delta)}{2^{a(1+\log^* \Delta)}} \right\rceil$$

admit a deterministic list edge-coloring in $O_a(\ell)$ rounds on general graphs. Thus these are a family of sufficient upper bounds, with constants depending on a .

Proof. Initially obtain a proper $O(\Delta^2)$ -vertex-coloring and a proper $s = O(\Delta)$ auxiliary edge-coloring in $O(\ell)$ rounds, using Lemma 7.2 and Theorem 7.1. Put $p = \lceil 2^{ah/2} \rceil$. For each fixed a , all sufficiently large Δ satisfy $p \leq \Delta/2$. Put $b = \lfloor \Delta/p \rfloor$ and compute a b -defective vertex-coloring with $O((\Delta/b)^2 \log(2\Delta/b)) = O(p^2 \log(2p))$ colors in $O(h)$ rounds by Lemma 7.2.

Apply Lemma 7.3 to these defective colors. It partitions the bichromatic edges into $O(\log p) = O_a(h)$ supplied-bipartite subgraphs, each of maximum degree at most Δ . Color these sequentially with Theorem 5.3. This is feasible for a sufficiently large C_a , since

$$\frac{\Delta^2 \log(2\Delta)}{2^{ah}} = \Delta^{2-o(1)} \gg \rho\Delta + \Delta,$$

and at most $2\Delta - 2$ colors can be removed from any list throughout the entire process. The residual graph has maximum degree at most b , consisting only of monochromatic edges of the defective vertex-coloring.

Its line graph has maximum degree at most $2b - 2$ and inherits the proper s -coloring of its vertices from the auxiliary edge-coloring. Furthermore,

$$b^2 \log(2b) \leq \frac{\Delta^2 \log(2\Delta)}{p^2} \leq \frac{\Delta^2 \log(2\Delta)}{2^{ah}}.$$

For sufficiently large C_a , the remaining lists therefore satisfy Lemma 7.6, including its $s = O(\Delta)$ requirement. Finish in constant time using that lemma on the line graph. Every color used is an actual member of the edge's original list.

The extra time beyond initialization is $O_a(h)$, and $h = O(\ell)$. All degree bounds excluded by “sufficiently large” form a finite set depending only on a . Increase C_a so that their lists have at least $2\Delta - 1$ elements and use a constant-degree greedy schedule after the initial symmetry breaking. This covers every $\Delta \geq 1$. $\square$

8 Lower bounds and limits of the constructions

8.1 The exact one-round threshold

Theorem 8.1. *Fix $\Delta \geq 2$ and a palette size q independent of n . Under our initialization and two-endpoint output conventions, any deterministic one-round algorithm valid on all sufficiently large Δ -regular supplied-bipartite graphs requires $q \geq \Delta^2$, even with identifiers and knowledge of n . The bound also holds for randomized algorithms with any fixed positive global success probability on all such graphs, if disconnected inputs are allowed. The port-pair algorithm attains Δ^2 .*

Deterministic case. Write $d = \Delta$ and use $H = K_{d,d}$ as a fixed core with a specified bipartition. Let P be a pool of K identifiers, where K is a multiple of $2d$, $K \geq 3d + 1$, and K is large enough for the finite Ramsey theorem with q^{d^2} edge colors and a homogeneous set of $d + 1$ vertices. There are finitely many core tests: all injective assignments of pool identifiers to H and all port numberings, in total

$$N = (K)_{2d}(d!)^{2d}.$$

Fix the global parameter n before considering these tests. For sufficiently large n divisible by $2d$, each test can be padded to that n with disjoint regular components. Padding is invisible in one round.

Suppose a deterministic protocol succeeds on every core test. For an edge whose black and white endpoint identifiers are x, y and whose ports are p, s , agreement forces the common output to be a function $f_n(x, y, p, s)$ of just these data. To prove this, hold the white endpoint's star fixed while varying the identifiers and opposite ports at the other neighbors of the black endpoint. These changes cannot affect the white endpoint's first-round view: first-round messages depend only on the sender's own initial information. For any two choices of the black star, choose one white star whose private identifiers avoid both choices. At most $3d - 1$ identifiers are needed. These compatible stars are realizable in $K_{d,d}$; complete unused vertices with distinct identifiers. The edges between private vertices add no initial information. Comparing through this common white star removes dependence on the black private star. The symmetric comparison removes the white private star, proving the claim without violating identifier uniqueness.

Color every unordered pool pair $x < y$ by the matrix $(f_n(x, y, p, s))_{p,s \in [d]}$. There are at most q^{d^2} matrices. Ramsey gives $d + 1$ identifiers on whose pairs the matrix is constant; denote it by $(a_{p,s})$. At a black center with the smallest identifier in this set, different local ports p, p' may have arbitrary opposite ports s, s' , forcing $a_{p,s} \neq a_{p',s'}$ by properness. At a white center with the largest identifier, different local ports s, s' may have the same opposite port p , forcing $a_{p,s} \neq a_{p,s'}$. All these stars occur among the tests. Therefore all d^2 entries differ, and $q \geq d^2$.

Only the elementary finite form of Ramsey's theorem is used. Its existence follows by successively choosing a vertex and retaining a largest color class among its edges to the unused vertices. For c colors, a sufficiently large finite starting set permits $c(k - 1) + 1$ such choices, each with a fixed color toward all later choices. At least k chosen vertices have the same fixed color and form a monochromatic clique. Thus K depends only on d, q , not on n or on the algorithm. $\square$

Randomized case. The preceding proof shows that every deterministic protocol with $q < d^2$ fails on at least one of the N core tests, for each fixed n . Take $n = tK$ and partition its identifiers into t disjoint pools of size K . In each pool independently choose a uniformly random core test and fill its unused identifiers by a fixed collection of copies of $K_{d,d}$. This defines a distribution on valid n -vertex inputs.

Fix all algorithmic randomness first: one public tape and a private tape for each identifier. In each pool the resulting deterministic algorithm fails at least one of its N tests. The preceding argument applies to any ordered identifier pool. Because input choices in different pools are independent and outputs in a component depend only on that component and the fixed global parameters,

$$\Pr_{\text{input}} [\text{global success} \mid \text{all tapes}] \leq (1 - 1/N)^t.$$

Average over the tapes. Some fixed input has algorithmic success probability at most $(1 - 1/N)^t$, tending to zero. This excludes any fixed positive global success guarantee. The argument uses independence of the input choices after conditioning on all tapes, not independence of component failures under public randomness. $\square$

The quantifiers fix d, q and allow arbitrarily large n ; the Ramsey threshold may be enormous. A separate promise $n = \text{poly}(d)$ is not covered. Nor does the proof apply unchanged if opposite ports or neighbor identifiers are supplied initially, or if only one endpoint must output. The randomized amplification uses disjoint unions. The theorem gives no superconstant round lower bound for a linear palette.

8.2 The small-palette obstruction

Chang et al. [12, Theorem 2] prove lower bounds $\Omega(\log_\Delta n)$ deterministically and $\Omega(\log_\Delta \log n)$ randomly for $(2\Delta - 2)$ -edge coloring, even with a supplied vertex bipartition. Its reduction explains the threshold: divide the colors into two groups of $\Delta - 1$ and orient their edges in opposite directions across the bipartition. Every degree- Δ vertex sees both groups, so the orientation has no sink. The published sinkless-orientation lower bound therefore applies, on high-girth regular instances. The relevant neighborhoods are trees; a finite Δ -regular tree is not needed.

For any fixed $\varepsilon > 0$, choose Δ large enough that $(2 - \varepsilon)\Delta \leq 2\Delta - 2$, and then let n grow. No round bound depending only on Δ can provide this palette uniformly over n . This leaves $2\Delta - 1$ and 2Δ untouched and gives no superlinear two-round palette lower bound. Without a supplied bipartition, the usual $\Omega(\log^* n)$ symmetry-breaking obstruction already occurs on cycles for a fixed finite palette [5]; hence the additive n -term in [table 2](#).

8.3 A limitation of fixed-fraction expansion

Proposition 8.2. *Let $M \geq \Delta$ left vertices have neighborhoods of common size $a \geq 1$ in a Q -element right universe. Suppose*

$$|N(I)| \geq \beta a |I| \quad (|I| \leq \Delta), \quad 0 < \beta \leq 1, \quad \beta \Delta \geq 2.$$

Then $Q > (e\beta/2)\Delta \ln(M/\Delta)$. In particular constant-fraction expansion requires $Q = \Omega(\Delta \log(M/\Delta))$.

Proof. Put $t = \lceil \beta a \Delta \rceil - 1$. Expansion gives $Q \geq \beta a \Delta$, so $a \leq t < Q$. Every right-side t -set contains the full neighborhoods of at most $\Delta - 1$ left vertices. Choose such a t -set T uniformly. For any particular left vertex,

$$\Pr[N_i \subseteq T] = \frac{\binom{t}{a}}{\binom{Q}{a}} \geq \left(\frac{t - a + 1}{Q} \right)^a \geq \left(\frac{\beta a \Delta}{2Q} \right)^a,$$

using $\beta \Delta \geq 2$. Linearity of expectation gives $M(\beta a \Delta / (2Q))^a \leq \Delta - 1 < \Delta$. Taking logarithms and maximizing $(\ln z)/z$ over positive z gives

$$\ln(M/\Delta) < a \ln \frac{2Q}{\beta a \Delta} \leq \frac{2Q}{e\beta \Delta}.$$

□

This applies to [lemma 2.2](#), whose expansion fraction exceeds $1/2$, for $\Delta \geq 4$. Its universe size is thus optimal in order under that expansion requirement. The sharper families of [theorem 3.5](#) need not arise from such neighborhoods.

8.4 A limitation of universal list richness

Proposition 8.3. *Suppose $C \geq 2r\Delta$ and there are Δ families of nonempty palettes in $[C]$, each palette of size at most r . Assume every $r\Delta$ -set contains a palette from each family, and every choice of one palette from each family has distinct representatives. Then*

$$(4r)^r \geq \frac{\Delta + 1}{\ln(2\Delta)}.$$

In particular $r = \Omega(\log \Delta / \log \log \Delta)$.

Proof. Fix a set W of $2r\Delta$ colors. An inclusion-maximal disjoint packing of palettes from one family inside W contains at least $\Delta + 1$ palettes. Otherwise its union has size at most $r\Delta$; its complement contains an $r\Delta$ -set and hence, by richness, another disjoint palette, a contradiction.

Include each color of W independently in a random set T with probability $1/(4r)$. Its expected size is $\Delta/2$, so $\Pr[|T| \geq \Delta] \leq 1/2$. Use $\Delta + 1$ disjoint packed palettes from each family. Each lies inside T with probability at least $(4r)^{-r}$, independently within that packing. The probability that T contains no palette from a given family is at most $\exp(-(\Delta + 1)/(4r)^r)$. If the claimed inequality failed, the union bound

$$\Pr[|T| \geq \Delta \text{ or some family has no palette in } T] \leq \frac{1}{2} + \Delta \exp\left(-\frac{\Delta + 1}{(4r)^r}\right) < 1$$

would give a set of fewer than Δ colors containing a palette from every family, contradicting distinct representatives. Taking logarithms yields $r \ln(4r) \geq \ln(\Delta + 1) - \ln \ln(2\Delta)$ and hence the assertion, indeed with leading lower bound $(1 - o(1)) \ln \Delta / \ln \ln \Delta$. $\square$

The table in [theorem 5.3](#) meets these hypotheses at white degree Δ , with $C = 2\rho\Delta^2$. Merely improving its probability estimates cannot make its palette-size parameter constant. This limits the specified universal richness and robustness requirements. Families adapted to more communicated list information, weaker compatibility conditions, or an iterative list-preserving reduction may avoid it.

9 What the results leave open

The elementary algorithm isolates a useful asymmetry: one endpoint coordinates proposals for its incident edges, and the other coordinates acceptance. Expansion makes the independent local decisions compatible. For vertex coloring there is no analogous division of constraints into the two endpoints of a variable; transferring this mechanism needs further work.

The main quantitative questions are constant-round ordinary $O(\Delta)$ coloring with a supplied bipartition, $O(h)$ rounds for arbitrary linear lists, and smaller final palette coefficients in $O(h)$ rounds. The exact two-round gap remains between a linear lower threshold and the $O(\Delta^{5/3})$ upper bound. The construction-specific barriers above do not resolve these distributed questions.

There is also a representational question. A palette family can encode an enormous collection of local proposals while admitting a short expansion or matching proof. This fits the round-elimination viewpoint [13], where the active/passive proposal step is one round. A symbolic treatment retaining useful relaxations of such families could help explain algorithms without enumerating their full finite descriptions. No additional algorithmic bound is asserted here.

A A proof of Haxell's matching condition

This is the bipartite hypergraph matching condition of Haxell [2, 3], with a hyperedge consisting of one family index and at most r universe elements. We include its alternating-sequence proof to keep all matching arguments self-contained.

Proof of Theorem 3.3. Induct on $n = |I|$, with $n = 0$ immediate. By induction, fix a matching M choosing disjoint palettes for all families except one root i_0 . A matching element retains its family label. We show how to augment M .

Maintain a sequence of candidate palettes $a_1, \dots, a_t$ and nonempty sets $B_1, \dots, B_t$ of matching elements, with these invariants:

1. The candidate palettes are pairwise disjoint.
2. B_j consists of all elements of M intersecting a_j , and the sets B_j are pairwise disjoint as sets of matching elements.
3. The family of a_j is the root or the family of a matching element in $B_1 \cup \dots \cup B_{j-1}$.

The empty sequence satisfies these conditions. Put $B = \bigcup_j B_j$, and let U be the union of the candidate palettes and all palettes in B . Each element of B intersects exactly its associated candidate; these intersections are disjoint because matching palettes are disjoint. Since $t \leq |B|$,

$$|U| \leq rt + r|B| - |B| \leq (2r - 1)|B|.$$

Let J contain the root and all families represented by B , so that $|J| = |B| + 1$. The hypothesis guarantees a palette a from a family in J that avoids U .

If a intersects an element of M , append a and all its blockers to the sequence. Its blockers are new, because a avoids U . A new blocker cannot intersect an earlier candidate, since all blockers of those candidates were already in B . Thus every invariant persists.

Otherwise a is unblocked. If its family is the root, insert a into M and finish. If not, its family has a unique matching element $b \in B_j$. Replace b in M by a , discard all sequence entries after j , and remove b from B_j . This is a valid matching swap, and earlier candidates avoid the new matching element a . If B_j is still nonempty, resume the procedure. If it is empty, remove entry j and treat a_j as the next unblocked palette. By the third invariant, its family is the root or has its matching element in an earlier blocker set. Repeat the same swap rule. This cascade moves strictly backwards, so it either finishes at the root or stops with an admissible sequence having only nonempty blocker sets. The earlier invariants remain valid throughout: a palette inserted into M avoids all earlier candidates, and no earlier candidate depends on the removed blocker; any later candidate that did depend on it was discarded.

To prove termination, associate with every admissible sequence the length- n integer vector

$$(|B_1|, \dots, |B_t|, n, \dots, n).$$

There are at most $n - 1$ matching elements, and $t \leq n - 1$. Appending a blocker set strictly decreases this vector in lexicographic order, because the new entry is at most $n - 1$. A swap and its backwards cascade, unless they finish, leave a prefix unchanged and decrease the blocker count at the next retained entry; truncation affects only later coordinates. They therefore strictly decrease the vector as well. Only finitely many such vectors exist. The procedure must finish the augmentation, proving the induction step. $\square$

B An explicit coefficient of 16.5

Write $d = \Delta$ throughout this appendix. The elementary proof of Theorem 2.4 gives $512d$ colors. The following refinement preserves its asymptotic round complexity. Its purpose is to record a concrete coefficient; no optimality is claimed.

We prove six constant-parameter palette reductions, as used in Theorem 3.7. In every row of Table 3, an input palette of size $\lfloor Ld \rfloor$ is replaced by one of size $\lfloor Rd \rfloor$. Every sampled candidate has exactly six elements.

Fix a row, and suppose $d \geq 6000$. Put

$$M = \lfloor Ld \rfloor, \quad Q = \lfloor Rd \rfloor, \quad S = R - \frac{1}{1000}, \quad \kappa = 11.$$

Independently sample m uniform six-element subsets of $[Q]$ for each of M families. Repetitions between samples are allowed. The choice of S handles both rounding and sampling without

Row	L	R	Candidates per family m
1	512	100	500
2	100	20	3500
3	20	18	6200
4	18	17	10400
5	17	83/5	10200
6	83/5	33/2	10500

Table 3: Six reductions with a common existence proof.

replacement:

$$Q - 5 \geq Rd - 6 \geq Sd. \quad (5)$$

There are two kinds of bad events. A failure of robust Hall yields k distinct families and a $(k - 1)$ -element set containing a candidate from each family, for some $1 \leq k \leq d$. The probability of such a witness, already summed over the choices of families and of the containing set, is at most

$$A_k \leq \binom{M}{k} \binom{Q}{k-1} \left[m \left(\frac{k}{Sd} \right)^6 \right]^k. \quad (6)$$

For $k = 1$ the event is impossible, but including its upper bound is harmless. For disjoint palette choices, we impose the stronger sufficient condition that no k families have all their candidates met by an $11k$ -element set, for any $1 \leq k \leq d$. This implies the hypothesis of Theorem 3.3: a smaller forbidden transversal can be enlarged because $Q \geq 11d$. Write

$$h(x) = 1 - \left(1 - \frac{11x}{S} \right)^6, \quad 0 < x \leq 1.$$

For a fixed set of size $11k$, the probability that a uniform six-element subset meets it is

$$1 - \prod_{j=0}^5 \left(1 - \frac{11k}{Q-j} \right) \leq h(k/d),$$

by (5). Thus the second bad-event probability is

$$B_k \leq \binom{M}{k} \binom{Q}{11k} h(k/d)^{mk}. \quad (7)$$

Here is a uniform bound on both sums. Let $H(z) = -z \log z - (1 - z) \log(1 - z)$ denote binary entropy, with natural logarithms, and define

$$\begin{aligned} F_H(x) &= RH(x/R) + LH(x/L) + x \log m + 6x \log(x/S), \\ F_C(x) &= LH(x/L) + RH(11x/R) + mx \log h(x), \\ p &= h(1), \quad \alpha = \frac{h'(1)}{h(1)}, \quad \beta = m\alpha - 12, \\ a &= \frac{e^2 L R m}{S^6}, \quad b = eL \left(\frac{eR}{11} \right)^{11} p^m. \end{aligned}$$

The following numerical bounds hold for all six rows:

$$F_H(1) < -\frac{9}{1000}, \quad F_C(1) < -\frac{3}{25}, \quad a < 2, \quad \beta > 160, \quad \log b < 6. \quad (8)$$

A rational certificate for these bounds is given below.

For completeness, the entropy estimate used here follows directly from the binomial theorem: for $0 < t < N$, the t th summand in $(t/N + (1 - t/N))^N = 1$ gives $\binom{N}{t} \leq \exp(NH(t/N))$; the endpoints follow by continuity. For fixed t , the function $uH(t/u)$ increases with u , since its derivative is $-\log(1 - t/u) > 0$. Hence the same upper bound is valid with any real upper bound on the population. Also $\binom{Q}{k-1} \leq \binom{Q}{k}$ for $k \leq d$, since $Q \geq 2d$. Applying these facts to (6) and (7) gives

$$A_k \leq e^{dF_H(k/d)}, \quad B_k \leq e^{dF_C(k/d)}. \quad (9)$$

Both ratios $F_H(x)/x$ and $F_C(x)/x$ increase on $(0, 1]$. Indeed, differentiation gives

$$\begin{aligned} x^2 \left(\frac{F_H(x)}{x} \right)' &= R \log(1 - x/R) + L \log(1 - x/L) + 6x, \\ x^2 \left(\frac{F_C(x)}{x} \right)' &= L \log(1 - x/L) + R \log(1 - 11x/R) + mx \frac{xh'(x)}{h(x)}. \end{aligned}$$

Writing $y = 1 - 11x/S$ shows that the elasticity

$$\frac{xh'(x)}{h(x)} = \frac{6}{1 + y^{-1} + \dots + y^{-5}}$$

is decreasing, and hence is at least α . The inequality $-\log(1 - u) \leq u/(1 - u)$, together with $L, R \geq 33/2$, bounds the two negative terms in the first derivative below by $-2x$ each. In the second derivative they are bounded below by $-2x$ and $-33x$, respectively. Since $m\alpha = \beta + 12 > 172$, both derivatives are positive. It follows from (8) and (9) that for $k > d/2$,

$$A_k \leq e^{-9k/1000}, \quad B_k \leq e^{-9k/1000}.$$

Each large-witness sum is therefore smaller than

$$\frac{e^{-27}}{1 - e^{-9/1000}} \leq \frac{1009}{9 \cdot 2^{27}} < \frac{1}{100}.$$

Here $d \geq 6000$, $e > 2$, and $1 - e^{-u} \geq u/(1 + u)$ were used.

Small witnesses require the factors depending on k/d . The bound $\binom{N}{t} \leq (eN/t)^t$ applied to the same counting formulas gives

$$A_k \leq \left[a(k/d)^4 \right]^k.$$

Integrating the elasticity inequality gives $h(x) \leq px^\alpha$, so

$$B_k \leq \left[b(k/d)^\beta \right]^k.$$

For $k \leq d/2$, the first bracket is less than $2/16 = 1/8$. The second is less than $e^6 2^{-160} < 1/8$. Each small-witness sum is at most $\sum_{k \geq 1} 8^{-k} = 1/7$. Consequently the probability of any obstruction is less than

$$\frac{2}{7} + \frac{2}{100} < 1.$$

A suitable table exists in every row. Theorem 3.3 and Lemma 3.2 turn it into the stated palette reduction.

Row	$F_H(1)$	$F_C(1)$	$A <$	$\beta >$
1	$(-8.578853, -8.578852)$	$(-301.671413, -301.671412)$	0.001	354.334
2	$(-0.243118, -0.243117)$	$(-9.810845, -9.810844)$	0.985	202.859
3	$(-0.777223, -0.777222)$	$(-5.473060, -5.473059)$	0.591	190.828
4	$(-0.084094, -0.084093)$	$(-5.209509, -5.209508)$	1.188	209.447
5	$(-0.044046, -0.044045)$	$(-0.619635, -0.619634)$	1.239	165.355
6	$(-0.009584, -0.009583)$	$(-0.121551, -0.121550)$	1.284	160.982

Table 4: Outward rational bounds for the six parameter rows.

Self-contained numerical certificate. All entries of Table 4 are rational bounds; a terminating recipe to check them follows. In its last two columns, $A = 9LRm/S^6$ is a rational upper bound on a because $e < 3$, and $\beta = m\alpha - 12$ is itself rational. The entries in the two middle columns are open intervals containing the indicated endpoint values.

For a positive rational number u , choose an integer j so that $1 \leq v = 2^{-j}u < 2$, and put $z = (v - 1)/(v + 1)$. With $N = 40$, define

$$P(v) = 2 \sum_{i=0}^{N-1} \frac{z^{2i+1}}{2i+1}, \quad E(v) = \frac{2z^{2N+1}}{(2N+1)(1-z^2)}.$$

The geometric expansion of $1/(1-z^2)$, integrated term by term on $[0, z]$, proves

$$P(v) \leq \log v \leq P(v) + E(v).$$

Apply the same formula at $v = 2$ to enclose $\log 2$, and use $\log u = j \log 2 + \log v$. When multiplying an interval by a negative number, interchange its endpoints. Evaluate H , $F_H(1)$, and $F_C(1)$ from their displayed formulas using interval addition and rational scalar multiplication. This produces the open enclosures in Table 4; every operation is rational, and the error formula specifies all discarded terms. The inequalities for A and β are direct rational substitutions.

The remaining estimate $\log b < 6$ does not need another numerical check. Expanding the entropy terms gives

$$\begin{aligned} \log b - F_C(1) &= 1 + (L - 1) \log(1 - 1/L) \\ &\quad + 11 + (R - 11) \log(1 - 11/R). \end{aligned}$$

The first line is less than $1/L \leq 2/33$. The function $11 + (R - 11) \log(1 - 11/R)$ decreases with R , since its derivative is $\log(1 - 11/R) + 11/R < 0$. At $R = 33/2$ it equals $11 - (11/2) \log 3 < 11/2$, because $e < 3$. Since $F_C(1) < 0$, these estimates imply $\log b < 6$ and complete (8).

C Elementary vertex-coloring tools

Lemma C.1 (Defective color reduction). *Let a graph of maximum degree at most $D \geq 1$ have a proper vertex M -coloring. For every real $p \geq 1$, a deterministic algorithm computes a vertex coloring with defect at most D/p and $O(p^2 \log(2p))$ colors in $O(1 + \log^* M)$ rounds. The constants in both bounds are absolute.*

Proof. We give a slightly less economical version of the color-reduction method of Kuhn [6, Section 4]; the extra logarithm in the palette will be irrelevant to our applications. A coloring has defect b if each vertex has at most b neighbors with its color.

We first prove a one-round reduction. Suppose the current coloring has $m \geq 2$ colors and defect at most b , and choose an integer $q \geq 1$. Put $a = \lceil 8q \ln m \rceil$. There exist functions

$f_i : [a] \rightarrow [4q]$, one for each old color $i \in [m]$, such that distinct functions agree at at most a/q arguments. Indeed, choose all their values independently and uniformly. For two fixed indices, the probability of agreement at more than a/q arguments, with $t = \lfloor a/q \rfloor + 1$, is zero if $t > a$, and otherwise at most

$$\binom{a}{t} (4q)^{-t} \leq \left(\frac{3a}{4qt} \right)^t \leq (3/4)^t \leq (3/4)^{8 \ln m}.$$

A union bound over fewer than m^2 pairs gives probability less than one, since $8 \ln(4/3) > 2$. Fix such a table by finite exhaustive search.

After receiving its neighbors' old colors, a vertex of old color i chooses an argument x minimizing the number of neighbors of color j for which $f_j(x) = f_i(x)$, and outputs $(x, f_i(x))$. Neighbors with old color i contribute at most b to this count at every argument. Averaged over the arguments, all other neighbors contribute at most D/q . Consequently the new defect is at most $b + D/q$. A neighbor with the same new color is necessarily included in the count at the chosen argument, even though it chooses its own argument independently. Writing $L = \log_2 m \geq 1$, the new number of colors is at most

$$4qa \leq 64q^2 L. \quad (10)$$

Let $K = 2^{15}$. Starting from the proper coloring, repeat this reduction while $m > (Kp)^{32}$, each time taking $q = \lceil 16p(\log_2 m)^2 \rceil$. The new palette bound m' satisfies

$$m' \leq Kp^2 L^5, \quad L' := \log_2 m' \leq A + 5 \log_2 L, \quad A = 15 + 2 \log_2 p.$$

Whenever a reduction is performed, $L > 32(15 + \log_2 p) \geq 480$. Thus $A < L/16$ and $\log_2 L \leq L/16$, whence $L' < L/2$. If the successive logarithms before these reductions are $L_0, \dots, L_{t-1}$, their reciprocal squares have sum at most $4/3$: read the sequence backwards and use $L_i > 2L_{i+1}$ and $L_{t-1} \geq 1$. The total increase in defect is therefore at most

$$\sum_{i < t} \frac{D}{16pL_i^2} \leq \frac{D}{12p}.$$

There are $O(1 + \log^* M)$ reductions, uniformly in p . To see this explicitly, if $5 \log_2 L \leq A$, then $L' \leq 2A < 32(15 + \log_2 p)$ and the loop stops. Every other step obeys $L' \leq 10 \log_2 L$. Above a fixed absolute threshold, two applications of $x \mapsto 10 \log_2 x$ are bounded by one application of $x \mapsto \log_2 x$. For example, for $x \geq 2^{1024}$ this follows from $10 \log_2(10z) \leq z$ for $z = \log_2 x \geq 1024$. Below that threshold, the preceding halving bound allows only an absolute constant number of further iterations.

At the end $m \leq (Kp)^{32}$. Apply the one-round reduction once more with $q = \lceil 2p \rceil$. Its defect contribution is at most $D/(2p)$, so the final defect is less than D/p . By (10), its palette has size

$$O(p^2 \log m) = O(p^2 \log(2p)).$$

If the initial proper coloring has at most one color, the graph is edgeless and requires no reduction. This completes the proof. $\square$

Corollary C.2 (Initial proper vertex coloring). *A graph of maximum degree at most $D \geq 1$, with a proper M -vertex coloring, admits a deterministic proper $O(D^2)$ -vertex coloring in $O(1 + \log^* M)$ rounds. In particular, polynomial-size unique identifiers on an n -vertex graph give such a coloring in $O(1 + \log^* n)$ rounds.*

Proof. This is Linial's color-reduction theorem [5]; the following derivation supplies the version needed here. Starting from the proper M -coloring, apply Lemma C.1 with $p = 2D$. The resulting defect is less than $1/2$, hence zero, and its palette has size at most $CD^2 \log(2D)$ for an absolute constant C . Choose a sufficiently large fixed constant c , and a prime $cD < q < 2cD$, as guaranteed by Bertrand's postulate. Taking c large enough ensures both $q > 2D$ and $q^3 \geq CD^2 \log(2D)$ for every integer $D \geq 1$. Encode the old colors as distinct polynomials of degree at most two over the field $\mathbb{F}_q$. After learning its neighbors' polynomials, each vertex chooses a field element x at which its polynomial differs from every neighbor's polynomial. Such an x exists, since each neighbor forbids at most two arguments and $q > 2D$. Output $(x, P(x))$. If adjacent vertices choose the same argument, their values differ; otherwise their first coordinates differ. This is therefore a proper $q^2 = O(D^2)$ -coloring, obtained in one additional round. For the identifier corollary take $M = n^{O(1)}$. $\square$

D Constant-time coloring from large lists

Lemma D.1 (Constant-round list coloring with large lists). *There is an absolute constant B such that a graph of maximum degree $D \geq 2$, supplied with a proper s -vertex-coloring, admits a deterministic list-coloring algorithm in constant time whenever every list has at least*

$$\max\{s, \lceil BD^2 \ln(2D) \rceil\}$$

colors. The color labels may be arbitrary finite strings. For example, $B = 2^{20}$ suffices.

Proof. We prove the needed special case of the conflict-coloring method of Fraigniaud, Heinrich, and Kosowski [10], with deliberately generous constants. A conflict-coloring instance has the common local list $[l]$ at each vertex. Each edge specifies forbidden pairs of local colors; for each color at either endpoint, at most b colors at the other endpoint are forbidden. Write $R = l/b$.

We first describe one simplification step, valid when $R \geq 96D^2$. Set

$$k = \left\lfloor \frac{l}{16bD} \right\rfloor, \quad t = \left\lfloor \frac{k}{D} \right\rfloor - 1, \quad N = \binom{l}{k}, \quad Q = \binom{kb}{t} \binom{l}{k-t}.$$

Candidates for the new local colors are the k -subsets of $[l]$. Declare candidates S and T at adjacent vertices incompatible if more than t elements of S conflict with at least one element of T , or more than t elements of T conflict with at least one element of S . For a fixed S , its elements conflict with at most kb colors at the other endpoint. Thus there are at most Q candidates T containing at least t such colors: choose t of these colors and then the remaining $k - t$ elements. Counting the other condition with the endpoints exchanged shows that the bipartite incompatibility graph between all N candidates at either endpoint has at most $2NQ$ edges.

At a vertex discard every candidate with more than $4DQ$ incompatibilities along any one incident edge. For each edge at most $N/(2D)$ candidates are discarded, so at least $N/2$ remain. Retain the first $l' = \lfloor N/2 \rfloor$ in lexicographic order and renumber them by $[l']$. Exchange these retained lists with neighbors. The new conflict instance has conflict degree at most $b' = 4DQ$; all of its local input, including the renumbered forbidden pairs, is obtained in one communication round from the previous instance.

Conversely, any solution to the new instance specifies a candidate S_v at each vertex. After one exchange of these candidates, vertex v chooses an element of S_v conflicting with no element of any neighbor's candidate. Each neighbor forbids at most t elements of S_v , and $Dt < k$, so such an element exists. The resulting colors solve the preceding conflict instance.

We record how much the list-to-conflict ratio improves. Our assumptions give $k \geq 4D$, $t \geq k/(2D)$, and $l - k \geq 15bDk$. Using $\binom{kb}{t} \leq (ekb/t)^t$ and $l' \geq N/3$, we obtain

$$\frac{l'}{b'} \geq \frac{1}{12D} \frac{\binom{l}{k}}{\binom{kb}{t} \binom{l}{k-t}} \geq \frac{1}{12D} \left(\frac{(l-k)t}{ek^2b} \right)^t \geq \frac{2^t}{16D}.$$

Indeed, the base in parentheses is at least $15/(2e) > 2$. Also $t \geq R/(16D^2) - 3 \geq R/(32D^2)$. Consequently each simplification guarantees

$$R' \geq \frac{1}{16D} \exp\left(\frac{R}{64D^2}\right). \quad (11)$$

Now truncate the original lists to the common length $l = \max\{s, \lceil 2^{20} D^2 \ln(2D) \rceil\}$. In one round exchange the ordered original lists and local port numbers. At each vertex replace its labels by their local ranks in $[l]$ and record, for each neighbor, the pairs of ranks representing equal actual colors. This is a conflict instance with $b = 1$. The original labels will only be used again to decode the eventual answer.

Apply the simplification twice, obtaining ratios R_1 and R_2 . Equation (11) gives

$$R_1 \geq l^8, \quad R_2 \geq \exp(l^6).$$

Here are elementary estimates covering all allowed parameters. The function $x/(64D^2) - 8 \ln x$ is increasing for $x \geq 2^{20} D^2 \ln(2D)$. At its left endpoint,

$$8 \ln x + \ln(16D) \leq 108 \ln(2D) < 2^{14} \ln(2D) = x/(64D^2),$$

where we used $\ln(2^{20}) \leq 10 \ln(2D)$, $\ln D \leq \ln(2D)$, $\ln \ln(2D) \leq \ln(2D)$, and $\ln(16D) \leq 4 \ln(2D)$. This proves the first bound. Since $D^2 \leq l$ and $l \geq 128$, $l^8/(64D^2) - \ln(16D) \geq l^6$, proving the second. In particular the hypotheses of both simplifications hold.

It remains to solve the twice-simplified instance without further communication. There are only finitely many possible normalized local inputs at this point. We give a bound that does not depend on the original color names. An initial normalized local input consists of its degree, its incident ports and opposite port numbers, and at most D Boolean $l \times l$ conflict matrices. Use D port slots, each storing a presence bit, an opposite-port number in $\lceil \log_2(D+1) \rceil$ bits, and an $l \times l$ conflict matrix. Since $l \geq 128$ and $D^2 \leq l$, this uses at most $2Dl^2$ bits. After two simplification rounds a local input is a deterministic function of a depth-two unfolded port-labelled neighborhood. Such a neighborhood has at most $2D^2$ vertices. Including its port incidences and the normalized initial inputs, it has a fixed-length encoding of $8D^3l^2$ bits, padding unused fields. Hence the set $\mathcal{I}$ of possible twice-simplified local inputs satisfies

$$|\mathcal{I}| \leq 2^{8D^3l^2}.$$

All vertices can enumerate a common finite superset of these inputs by enumerating such neighborhood encodings, performing the two local transformations, and discarding malformed encodings, encodings with initial conflict degree greater than one, and encodings for which a required retained list is too short. Retain the resulting local inputs whose conflict matrices obey the prescribed conflict-degree bound. Every actual input survives these checks.

Pair each type in $\mathcal{I}$ with a proper vertex color in $[s]$. In a fixed order, assign an output from the common twice-simplified list to every such pair. When assigning a pair, avoid any color that conflicts, along any local port, with the output already assigned to any earlier pair. Each earlier pair excludes at most Db_2 colors, where b_2 is the new conflict-degree bound. This greedy table exists because

$$Ds|\mathcal{I}| < \exp(l^6) \leq R_2 = l_2/b_2.$$

For the strict inequality, use $s \leq l$, $D^2 \leq l$, and $l \geq 128$: the logarithm of its left side is at most $2 \ln l + 8D^3 l^2 \ln 2 \leq 9l^4 < l^6$. Adjacent vertices have different proper vertex colors, hence different type-color pairs. The later pair in the fixed order avoids the earlier pair's output on their actual edge. Looking up one's pair therefore gives a valid conflict-coloring without communication.

Finally invert the two simplifications, one round each, and translate local ranks back to actual input colors. This uses a constant number of communication rounds and always returns a color in the original list. The construction is deterministic: all tables are finite and chosen canonically, with unrestricted local computation. $\square$

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